

## **BDA Practical No.1**

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**Note:-**Ensure that Hadoop is installed in your system by typing the command given below after opening Ubuntu Terminal

hadoop version

If hadoop is installed then the output will show version number of Hadoop and if Hadoop is not installed then you will have to install Hadoop first in your system. You can refer my pdf named “Ubuntu and Hadoop Installation on Windows by Aditya Nikam” pdf.

- If Hadoop is installed: You will see a block of text telling you the version number (e.g., Hadoop 3.3.6) after typing the hadoop version given above
- If Hadoop is NOT installed: You will see an error message like hadoop: command not found

**Step1:-**Open Ubuntu terminal and Type the command given below:

start-dfs.sh

**If you get output error like given below:-**

aditya@DESKTOP-4CC1PF9\$ start-dfs.sh

Starting namenodes on [DESKTOP-4CC1PF9]

DESKTOP-4CC1PF9: Warning: Permanently added 'desktop-4cc1pf9' (ED25519) to the list of known hosts.

DESKTOP-4CC1PF9: aditya@desktop-4cc1pf9: Permission denied (publickey,password).

Starting datanodes

localhost: Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.

localhost: aditya@localhost: Permission denied (publickey,password).

Starting secondary namenodes [DESKTOP-4CC1PF9]

DESKTOP-4CC1PF9: aditya@desktop-4cc1pf9: Permission denied (publickey,password).

**Then type the below command:-**

```
ssh-keygen -t rsa -P "" -f ~/.ssh/id_rsa
```

**You may get output like this after typing this above command:-**

```
Generating public/private rsa key pair.
Your identification has been saved in /home/aditya/.ssh/id_rsa
Your public key has been saved in /home/aditya/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:8gOxRdX3g27UnRVO5hsSTbJhon0DzrFRaY5PzduOPLQ aditya@DESKTOP-
4CC1PF9
The key's randomart image is:
+---[RSA 3072]-----+
|    ...*. *+=. |
|    . = Oo@.. |
|    . .. =+*==+ |
|    + ..=o*= |
|    + S = .+ |
|    +  +... |
|    o .o + |
|    . E . |
|    . |
+----[SHA256]-----+
aditya@DESKTOP-4CC1PF9:~$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
aditya@DESKTOP-4CC1PF9:~$ chmod 0600 ~/.ssh/authorized_keys
aditya@DESKTOP-4CC1PF9:~$ sudo service ssh start
aditya@DESKTOP-4CC1PF9:~$ start-dfs.sh
Starting namenodes on [DESKTOP-4CC1PF9]
DESKTOP-4CC1PF9: ERROR: JAVA_HOME is not set and could not be found.
Starting datanodes
localhost: ERROR: JAVA_HOME is not set and could not be found.
Starting secondary namenodes [DESKTOP-4CC1PF9]
DESKTOP-4CC1PF9: ERROR: JAVA_HOME is not set and could not be found.
aditya@DESKTOP-4CC1PF9:~$
```

**Now type below command if you get above output:-**

```
nano /usr/local/hadoop/etc/hadoop/hadoop-env.sh
```

- **Now Use your Down Arrow Key to scroll down about 20-30 lines.**

- Look for a line that looks like # export JAVA\_HOME=.
- Right below it (or anywhere on a new line), paste this exact line given below:

export JAVA\_HOME=/usr/lib/jvm/java-8-openjdk-amd64 **(Dont put hasthag(#) at the starting)**

Now do Ctrl +O,then press Enter and then press Ctrl +X

**Now type:-**

start-dfs.sh

**If you get an error like given below:-**

```
Starting namenodes on [DESKTOP-4CC1PF9]
DESKTOP-4CC1PF9: WARNING: /usr/local/hadoop/logs does not exist. Creating.
DESKTOP-4CC1PF9: ERROR: Cannot set priority of namenode process 2012
Starting datanodes
Starting secondary namenodes [DESKTOP-4CC1PF9]
DESKTOP-4CC1PF9: ERROR: Cannot set priority of secondarynamenode process 2315
```

**Then type the below command:-**

jps

**You may see output something like given below:-**

```
3455 Jps
aditya@DESKTOP-4CC1PF9:~$
```

**Now type:-**

start-yarn.sh

**You may see output like below:-**

```
Starting resourcemanager
Starting nodemanagers
```

**Now type:-**

jps

**You will see Output like given below:-**

```
4309 NodeManager
4183 ResourceManager
4638 Jps
```

**Now type:-**

stop-all.sh

**You will get output like below:-**

```
WARNING: Stopping all Apache Hadoop daemons as aditya in 10 seconds.
WARNING: Use CTRL-C to abort.
Stopping namenodes on [DESKTOP-4CC1PF9]
Stopping datanodes
Stopping secondary namenodes [DESKTOP-4CC1PF9]
Stopping nodemanagers
Stopping resourcemanager
```

**Type:-**

hdfs namenode -format

**You will see output like given below:-**

```
2026-05-07 17:55:34,904 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG: host = DESKTOP-4CC1PF9/127.0.1.1
STARTUP_MSG: args = [-format]
STARTUP_MSG: version = 3.3.6
STARTUP_MSG: classpath =
/usr/local/hadoop/etc/hadoop:/usr/local/hadoop/share/hadoop/common/lib/woodstox-
core-5.4.0.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-codec-dns-
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/curator-client-
5.2.0.jar:/usr/local/hadoop/share/hadoop/common/lib/hadoop-annotations-
3.3.6.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-codec-haproxy-
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/jsr305-
3.0.2.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-resolver-dns-native-macos-
4.1.89.Final-osx-x86_64.jar:/usr/local/hadoop/share/hadoop/common/lib/jcip-
annotations-1.0-1.jar:/usr/local/hadoop/share/hadoop/common/lib/stax2-api-
4.2.1.jar:/usr/local/hadoop/share/hadoop/common/lib/kerb-identity-
1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/jackson-databind-
2.12.7.1.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-buffer-
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/hadoop-auth-
3.3.6.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-util-
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/common/lib/reload4j-
```

1.2.22.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-transport-native-epoll-  
4.1.89.Final-linux-x86\_64.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-codec-  
mqtt-4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-security-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/common/lib/kerb-core-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/kerb-crypto-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/paranamer-  
2.3.jar:/usr/local/hadoop/share/hadoop/common/lib/kerby-config-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/failureaccess-  
1.0.jar:/usr/local/hadoop/share/hadoop/common/lib/jsch-  
0.1.55.jar:/usr/local/hadoop/share/hadoop/common/lib/curator-recipes-  
5.2.0.jar:/usr/local/hadoop/share/hadoop/common/lib/httpclient-  
4.5.13.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-codec-redis-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-text-  
1.10.0.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-logging-  
1.1.3.jar:/usr/local/hadoop/share/hadoop/common/lib/audience-annotations-  
0.5.0.jar:/usr/local/hadoop/share/hadoop/common/lib/httpcore-  
4.4.13.jar:/usr/local/hadoop/share/hadoop/common/lib/zookeeper-jute-  
3.6.3.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-resolver-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-collections-  
3.2.2.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-codec-socks-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/token-provider-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-beanutils-  
1.9.4.jar:/usr/local/hadoop/share/hadoop/common/lib/nimbus-jose-jwt-  
9.8.1.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-handler-proxy-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-net-  
3.9.0.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-http-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/common/lib/kerby-pkix-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/jakarta.activation-api-  
1.2.1.jar:/usr/local/hadoop/share/hadoop/common/lib/kerb-server-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-codec-http2-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/kerb-admin-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-compress-  
1.21.jar:/usr/local/hadoop/share/hadoop/common/lib/jackson-mapper-asl-  
1.9.13.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-transport-native-kqueue-  
4.1.89.Final-osx-aarch\_64.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-  
lang3-3.12.0.jar:/usr/local/hadoop/share/hadoop/common/lib/kerb-util-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-codec-smtp-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-math3-

3.1.1.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-transport-sctp-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/jackson-annotations-  
2.12.7.jar:/usr/local/hadoop/share/hadoop/common/lib/hadoop-shaded-protobuf\_3\_7-  
1.1.1.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-configuration2-  
2.8.0.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-resolver-dns-native-macos-  
4.1.89.Final-osx-aarch\_64.jar:/usr/local/hadoop/share/hadoop/common/lib/protobuf-  
java-2.5.0.jar:/usr/local/hadoop/share/hadoop/common/lib/jersey-core-  
1.19.4.jar:/usr/local/hadoop/share/hadoop/common/lib/javax.servlet-api-  
3.1.0.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-codec-  
1.15.jar:/usr/local/hadoop/share/hadoop/common/lib/dnsjava-  
2.1.7.jar:/usr/local/hadoop/share/hadoop/common/lib/zookeeper-  
3.6.3.jar:/usr/local/hadoop/share/hadoop/common/lib/kerby-util-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/kerby-xdr-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-servlet-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-all-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/kerby-asn1-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-transport-rxtx-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-util-ajax-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-transport-udt-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/snappy-java-  
1.1.8.2.jar:/usr/local/hadoop/share/hadoop/common/lib/hadoop-shaded-guava-  
1.1.1.jar:/usr/local/hadoop/share/hadoop/common/lib/animal-sniffer-annotations-  
1.17.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-transport-classes-epoll-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/jettison-  
1.5.4.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-transport-native-kqueue-  
4.1.89.Final-osx-x86\_64.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-  
transport-classes-kqueue-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-transport-native-unix-  
common-4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-codec-xml-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/slf4j-api-  
1.7.36.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-handler-ssl-ocsp-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-transport-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-codec-http-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-common-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/curator-framework-  
5.2.0.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-io-  
2.8.0.jar:/usr/local/hadoop/share/hadoop/common/lib/guava-27.0-  
jre.jar:/usr/local/hadoop/share/hadoop/common/lib/jersey-servlet-

1.19.4.jar:/usr/local/hadoop/share/hadoop/common/lib/jaxb-impl-2.2.3-  
1.jar:/usr/local/hadoop/share/hadoop/common/lib/j2objc-annotations-  
1.1.jar:/usr/local/hadoop/share/hadoop/common/lib/listenablefuture-9999.0-empty-to-  
avoid-conflict-with-guava.jar:/usr/local/hadoop/share/hadoop/common/lib/metrics-core-  
3.2.4.jar:/usr/local/hadoop/share/hadoop/common/lib/jackson-core-  
2.12.7.jar:/usr/local/hadoop/share/hadoop/common/lib/avro-  
1.7.7.jar:/usr/local/hadoop/share/hadoop/common/lib/jersey-json-  
1.20.jar:/usr/local/hadoop/share/hadoop/common/lib/jul-to-slf4j-  
1.7.36.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-cli-  
1.2.jar:/usr/local/hadoop/share/hadoop/common/lib/jsr311-api-  
1.1.1.jar:/usr/local/hadoop/share/hadoop/common/lib/kerb-simplekdc-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/re2j-  
1.1.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-daemon-  
1.0.13.jar:/usr/local/hadoop/share/hadoop/common/lib/jsp-api-  
2.1.jar:/usr/local/hadoop/share/hadoop/common/lib/kerb-client-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/slf4j-reload4j-  
1.7.36.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-io-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-codec-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/jersey-server-  
1.19.4.jar:/usr/local/hadoop/share/hadoop/common/lib/checker-qual-  
2.5.2.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-server-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-codec-stomp-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-codec-memcache-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-webapp-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-resolver-dns-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-xml-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-resolver-dns-  
classes-macos-4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-  
handler-4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/gson-  
2.9.0.jar:/usr/local/hadoop/share/hadoop/common/lib/jaxb-api-  
2.2.11.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-transport-native-epoll-  
4.1.89.Final-linux-aarch\_64.jar:/usr/local/hadoop/share/hadoop/common/lib/jackson-  
core-asl-1.9.13.jar:/usr/local/hadoop/share/hadoop/common/lib/kerb-common-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/common/hadoop-nfs-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/common/hadoop-registry-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/common/hadoop-kms-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/common/hadoop-common-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/common/hadoop-common-3.3.6-

tests.jar:/usr/local/hadoop/share/hadoop/hdfs:/usr/local/hadoop/share/hadoop/hdfs/lib/  
woodstox-core-5.4.0.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-codec-dns-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/curator-client-  
5.2.0.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/json-simple-  
1.1.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/hadoop-annotations-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-codec-haproxy-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jsr305-  
3.0.2.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-resolver-dns-native-macos-  
4.1.89.Final-osx-x86\_64.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jcip-annotations-  
1.0-1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/stax2-api-  
4.2.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/kerb-identity-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jackson-databind-  
2.12.7.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-buffer-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/hadoop-auth-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jetty-util-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/reload4j-  
1.2.22.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-transport-native-epoll-  
4.1.89.Final-linux-x86\_64.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-codec-mqtt-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jetty-security-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/kerb-core-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/kerb-crypto-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/paranamer-  
2.3.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/kerby-config-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/failureaccess-  
1.0.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jsch-  
0.1.55.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/curator-recipes-  
5.2.0.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/httpclient-  
4.5.13.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-codec-redis-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/commons-text-  
1.10.0.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/commons-logging-  
1.1.3.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/audience-annotations-  
0.5.0.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/httpcore-  
4.4.13.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/zookeeper-jute-  
3.6.3.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-resolver-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/commons-collections-  
3.2.2.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-codec-socks-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/token-provider-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/commons-beanutils-



1.9.4.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/nimbus-jose-jwt-  
9.8.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-handler-proxy-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/commons-net-  
3.9.0.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jetty-http-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/kerby-pkix-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jakarta.activation-api-  
1.2.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/kerb-server-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-codec-http2-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/kerb-admin-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/commons-compress-  
1.21.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jackson-mapper-asl-  
1.9.13.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-transport-native-kqueue-  
4.1.89.Final-osx-aarch\_64.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/commons-lang3-  
3.12.0.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/kerb-util-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-codec-smtp-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/commons-math3-  
3.1.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-transport-sctp-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jackson-annotations-  
2.12.7.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/hadoop-shaded-protobuf\_3\_7-  
1.1.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/commons-configuration2-  
2.8.0.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-resolver-dns-native-macos-  
4.1.89.Final-osx-aarch\_64.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/protobuf-java-  
2.5.0.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jersey-core-  
1.19.4.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/javax.servlet-api-  
3.1.0.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/commons-codec-  
1.15.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/dnsjava-  
2.1.7.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/zookeeper-  
3.6.3.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/kerby-util-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/kerby-xdr-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jetty-servlet-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-all-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/kerby-asn1-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-transport-rxtx-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jetty-util-ajax-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-transport-udt-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/snappy-java-  
1.1.8.2.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/hadoop-shaded-guava-  
1.1.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/okio-

2.8.0.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/animal-sniffer-annotations-  
1.17.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-transport-classes-epoll-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jettison-  
1.5.4.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-transport-native-kqueue-  
4.1.89.Final-osx-x86\_64.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/okhttp-  
4.9.3.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-transport-classes-kqueue-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-transport-native-unix-  
common-4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-codec-xml-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-handler-ssl-ocsp-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-transport-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-codec-http-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-common-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/curator-framework-  
5.2.0.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/commons-io-  
2.8.0.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/guava-27.0-  
jre.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jersey-servlet-  
1.19.4.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jaxb-impl-2.2.3-  
1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/j2objc-annotations-  
1.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/listenablefuture-9999.0-empty-to-avoid-  
conflict-with-guava.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/HikariCP-java7-  
2.4.12.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/metrics-core-  
3.2.4.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jackson-core-  
2.12.7.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/kotlin-stdlib-  
1.4.10.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/avro-  
1.7.7.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-  
3.10.6.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jersey-json-  
1.20.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/commons-cli-  
1.2.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jsr311-api-  
1.1.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/kerb-simplekdc-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/re2j-  
1.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/commons-daemon-  
1.0.13.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/kerb-client-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jetty-io-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-codec-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jersey-server-  
1.19.4.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/checker-qual-  
2.5.2.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jetty-server-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-codec-stomp-

4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-codec-memcache-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jetty-webapp-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-resolver-dns-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jetty-xml-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-resolver-dns-  
classes-macos-4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/kotlin-stdlib-  
common-1.4.10.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-handler-  
4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/gson-  
2.9.0.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jaxb-api-  
2.2.11.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/netty-transport-native-epoll-  
4.1.89.Final-linux-aarch\_64.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/jackson-core-  
asl-1.9.13.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/kerb-common-  
1.0.1.jar:/usr/local/hadoop/share/hadoop/hdfs/lib/leveldbjni-all-  
1.8.jar:/usr/local/hadoop/share/hadoop/hdfs/hadoop-hdfs-client-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/hdfs/hadoop-hdfs-3.3.6-  
tests.jar:/usr/local/hadoop/share/hadoop/hdfs/hadoop-hdfs-native-client-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/hdfs/hadoop-hdfs-native-client-3.3.6-  
tests.jar:/usr/local/hadoop/share/hadoop/hdfs/hadoop-hdfs-rbf-3.3.6-  
tests.jar:/usr/local/hadoop/share/hadoop/hdfs/hadoop-hdfs-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/hdfs/hadoop-hdfs-nfs-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/hdfs/hadoop-hdfs-client-3.3.6-  
tests.jar:/usr/local/hadoop/share/hadoop/hdfs/hadoop-hdfs-rbf-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/hdfs/hadoop-hdfs-https-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-client-shuffle-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-client-core-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-client-  
jobclient-3.3.6.jar:/usr/local/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-  
client-jobclient-3.3.6-tests.jar:/usr/local/hadoop/share/hadoop/mapreduce/hadoop-  
mapreduce-client-common-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-client-app-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-client-  
uploader-3.3.6.jar:/usr/local/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-  
client-nativetask-3.3.6.jar:/usr/local/hadoop/share/hadoop/mapreduce/hadoop-  
mapreduce-client-hs-3.3.6.jar:/usr/local/hadoop/share/hadoop/mapreduce/hadoop-  
mapreduce-examples-3.3.6.jar:/usr/local/hadoop/share/hadoop/mapreduce/hadoop-  
mapreduce-client-hs-plugins-  
3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn:/usr/local/hadoop/share/hadoop/yarn/lib/  
websocket-common-

9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/yarn/lib/asm-commons-  
9.4.jar:/usr/local/hadoop/share/hadoop/yarn/lib/java-util-  
1.9.0.jar:/usr/local/hadoop/share/hadoop/yarn/lib/mssql-jdbc-  
6.2.1.jre7.jar:/usr/local/hadoop/share/hadoop/yarn/lib/aopalliance-  
1.0.jar:/usr/local/hadoop/share/hadoop/yarn/lib/snakeyaml-  
2.0.jar:/usr/local/hadoop/share/hadoop/yarn/lib/jetty-plus-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/yarn/lib/javax.inject-  
1.jar:/usr/local/hadoop/share/hadoop/yarn/lib/guice-  
4.0.jar:/usr/local/hadoop/share/hadoop/yarn/lib/websocket-api-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/yarn/lib/javax-websocket-client-  
impl-9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/yarn/lib/bcprov-jdk15on-  
1.68.jar:/usr/local/hadoop/share/hadoop/yarn/lib/swagger-annotations-  
1.5.4.jar:/usr/local/hadoop/share/hadoop/yarn/lib/jna-  
5.2.0.jar:/usr/local/hadoop/share/hadoop/yarn/lib/bcpkix-jdk15on-  
1.68.jar:/usr/local/hadoop/share/hadoop/yarn/lib/websocket-servlet-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/yarn/lib/websocket-server-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/yarn/lib/javax.websocket-client-  
api-1.0.jar:/usr/local/hadoop/share/hadoop/yarn/lib/javax-websocket-server-impl-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/yarn/lib/json-io-  
2.5.1.jar:/usr/local/hadoop/share/hadoop/yarn/lib/jackson-module-jaxb-annotations-  
2.12.7.jar:/usr/local/hadoop/share/hadoop/yarn/lib/jersey-client-  
1.19.4.jar:/usr/local/hadoop/share/hadoop/yarn/lib/jetty-jndi-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/yarn/lib/jline-  
3.9.0.jar:/usr/local/hadoop/share/hadoop/yarn/lib/jackson-jaxrs-json-provider-  
2.12.7.jar:/usr/local/hadoop/share/hadoop/yarn/lib/asm-tree-  
9.4.jar:/usr/local/hadoop/share/hadoop/yarn/lib/jetty-annotations-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/yarn/lib/javax.websocket-api-  
1.0.jar:/usr/local/hadoop/share/hadoop/yarn/lib/websocket-client-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/yarn/lib/jakarta.xml.bind-api-  
2.3.2.jar:/usr/local/hadoop/share/hadoop/yarn/lib/jetty-client-  
9.4.51.v20230217.jar:/usr/local/hadoop/share/hadoop/yarn/lib/guice-servlet-  
4.0.jar:/usr/local/hadoop/share/hadoop/yarn/lib/geronimo-jcache\_1.0\_spec-1.0-alpha-  
1.jar:/usr/local/hadoop/share/hadoop/yarn/lib/objenesis-  
2.6.jar:/usr/local/hadoop/share/hadoop/yarn/lib/ehcache-  
3.3.1.jar:/usr/local/hadoop/share/hadoop/yarn/lib/fst-  
2.50.jar:/usr/local/hadoop/share/hadoop/yarn/lib/jackson-jaxrs-base-  
2.12.7.jar:/usr/local/hadoop/share/hadoop/yarn/lib/jersey-guice-  
1.19.4.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-server-

```
sharedcachemanager-3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-api-
3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-services-api-
3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-server-tests-
3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-applications-mawo-core-
3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-server-router-
3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-registry-
3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-services-core-
3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-client-
3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-server-nodemanager-
3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-server-timeline-
pluginstorage-3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-applications-
distributedshell-3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-
applications-unmanaged-am-launcher-
3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-server-web-proxy-
3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-server-resourcemanager-
3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-server-common-
3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-common-
3.3.6.jar:/usr/local/hadoop/share/hadoop/yarn/hadoop-yarn-server-
applicationhistoryservice-3.3.6.jar
STARTUP_MSG: build = https://github.com/apache/hadoop.git -r
1be78238728da9266a4f88195058f08fd012bf9c; compiled by 'ubuntu' on 2023-06-
18T08:22Z
STARTUP_MSG: java = 1.8.0_482
*****/
2026-05-07 17:55:34,912 INFO namenode.NameNode: registered UNIX signal handlers for
[TERM, HUP, INT]
2026-05-07 17:55:34,968 INFO namenode.NameNode: createNameNode [-format]
2026-05-07 17:55:35,223 INFO namenode.NameNode: Formatting using clusterid: CID-
a26ab32e-129d-4700-86cd-739b5db8902b
2026-05-07 17:55:35,251 INFO namenode.FSEditLog: Edit logging is async:true
2026-05-07 17:55:35,274 INFO namenode.FSNamesystem: KeyProvider: null
2026-05-07 17:55:35,275 INFO namenode.FSNamesystem: fsLock is fair: true
2026-05-07 17:55:35,276 INFO namenode.FSNamesystem: Detailed lock hold time
metrics enabled: false
2026-05-07 17:55:35,279 INFO namenode.FSNamesystem: fsOwner          = aditya
(auth:SIMPLE)
2026-05-07 17:55:35,279 INFO namenode.FSNamesystem: supergroup      =
supergroup
```

2026-05-07 17:55:35,280 INFO namenode.FSNamesystem: isPermissionEnabled = true  
2026-05-07 17:55:35,280 INFO namenode.FSNamesystem: isStoragePolicyEnabled = true  
2026-05-07 17:55:35,280 INFO namenode.FSNamesystem: HA Enabled: false  
2026-05-07 17:55:35,313 INFO common.Util:  
dfs.datanode.fileio.profiling.sampling.percentage set to 0. Disabling file IO profiling  
2026-05-07 17:55:35,397 INFO blockmanagement.DatanodeManager:  
dfs.block.invalidate.limit : configured=1000, counted=60, effected=1000  
2026-05-07 17:55:35,397 INFO blockmanagement.DatanodeManager:  
dfs.namenode.datanode.registration.ip-hostname-check=true  
2026-05-07 17:55:35,400 INFO blockmanagement.BlockManager:  
dfs.namenode.startup.delay.block.deletion.sec is set to 000:00:00:00.000  
2026-05-07 17:55:35,400 INFO blockmanagement.BlockManager: The block deletion will  
start around 2026 May 07 17:55:35  
2026-05-07 17:55:35,402 INFO util.GSet: Computing capacity for map BlocksMap  
2026-05-07 17:55:35,402 INFO util.GSet: VM type = 64-bit  
2026-05-07 17:55:35,402 INFO util.GSet: 2.0% max memory 1.7 GB = 34.5 MB  
2026-05-07 17:55:35,402 INFO util.GSet: capacity =  $2^{22}$  = 4194304 entries  
2026-05-07 17:55:35,407 INFO blockmanagement.BlockManager: Storage policy satisfier  
is disabled  
2026-05-07 17:55:35,407 INFO blockmanagement.BlockManager:  
dfs.block.access.token.enable = false  
2026-05-07 17:55:35,411 INFO blockmanagement.BlockManagerSafeMode:  
dfs.namenode.safemode.threshold-pct = 0.999  
2026-05-07 17:55:35,411 INFO blockmanagement.BlockManagerSafeMode:  
dfs.namenode.safemode.min.datanodes = 0  
2026-05-07 17:55:35,411 INFO blockmanagement.BlockManagerSafeMode:  
dfs.namenode.safemode.extension = 30000  
2026-05-07 17:55:35,411 INFO blockmanagement.BlockManager: defaultReplication =  
3  
2026-05-07 17:55:35,411 INFO blockmanagement.BlockManager: maxReplication =  
512  
2026-05-07 17:55:35,411 INFO blockmanagement.BlockManager: minReplication = 1  
2026-05-07 17:55:35,411 INFO blockmanagement.BlockManager: maxReplicationStreams  
= 2  
2026-05-07 17:55:35,411 INFO blockmanagement.BlockManager:  
redundancyRecheckInterval = 3000ms  
2026-05-07 17:55:35,412 INFO blockmanagement.BlockManager: encryptDataTransfer  
= false



2026-05-07 17:55:35,412 INFO blockmanagement.BlockManager: maxNumBlocksToLog = 1000

2026-05-07 17:55:35,433 INFO namenode.FSDirectory: GLOBAL serial map: bits=29 maxEntries=536870911

2026-05-07 17:55:35,433 INFO namenode.FSDirectory: USER serial map: bits=24 maxEntries=16777215

2026-05-07 17:55:35,433 INFO namenode.FSDirectory: GROUP serial map: bits=24 maxEntries=16777215

2026-05-07 17:55:35,433 INFO namenode.FSDirectory: XATTR serial map: bits=24 maxEntries=16777215

2026-05-07 17:55:35,444 INFO util.GSet: Computing capacity for map INodeMap

2026-05-07 17:55:35,444 INFO util.GSet: VM type = 64-bit

2026-05-07 17:55:35,445 INFO util.GSet: 1.0% max memory 1.7 GB = 17.3 MB

2026-05-07 17:55:35,445 INFO util.GSet: capacity =  $2^{21}$  = 2097152 entries

2026-05-07 17:55:35,471 INFO namenode.FSDirectory: ACLs enabled? true

2026-05-07 17:55:35,471 INFO namenode.FSDirectory: POSIX ACL inheritance enabled? true

2026-05-07 17:55:35,471 INFO namenode.FSDirectory: XAttrs enabled? true

2026-05-07 17:55:35,471 INFO namenode.NameNode: Caching file names occurring more than 10 times

2026-05-07 17:55:35,475 INFO snapshot.SnapshotManager: Loaded config captureOpenFiles: false, skipCaptureAccessTimeOnlyChange: false, snapshotDiffAllowSnapRootDescendant: true, maxSnapshotLimit: 65536

2026-05-07 17:55:35,476 INFO snapshot.SnapshotManager: SkipList is disabled

2026-05-07 17:55:35,480 INFO util.GSet: Computing capacity for map cachedBlocks

2026-05-07 17:55:35,480 INFO util.GSet: VM type = 64-bit

2026-05-07 17:55:35,480 INFO util.GSet: 0.25% max memory 1.7 GB = 4.3 MB

2026-05-07 17:55:35,480 INFO util.GSet: capacity =  $2^{19}$  = 524288 entries

2026-05-07 17:55:35,485 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.window.num.buckets = 10

2026-05-07 17:55:35,485 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.num.users = 10

2026-05-07 17:55:35,485 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.windows.minutes = 1,5,25

2026-05-07 17:55:35,488 INFO namenode.FSNamesystem: Retry cache on namenode is enabled

2026-05-07 17:55:35,488 INFO namenode.FSNamesystem: Retry cache will use 0.03 of total heap and retry cache entry expiry time is 600000 millis

2026-05-07 17:55:35,489 INFO util.GSet: Computing capacity for map  
NameNodeRetryCache  
2026-05-07 17:55:35,489 INFO util.GSet: VM type = 64-bit  
2026-05-07 17:55:35,489 INFO util.GSet: 0.029999999329447746% max memory 1.7 GB =  
530.4 KB  
2026-05-07 17:55:35,489 INFO util.GSet: capacity =  $2^{16}$  = 65536 entries  
Re-format filesystem in Storage Directory root= /tmp/hadoop-aditya/dfs/name; location=  
null ? (Y or N) Y  
2026-05-07 17:55:37,583 INFO namenode.FSImage: Allocated new BlockPoolId: BP-  
1730856012-127.0.1.1-1778176537579  
2026-05-07 17:55:37,583 INFO common.Storage: Will remove files: [/tmp/hadoop-  
aditya/dfs/name/current/VERSION, /tmp/hadoop-  
aditya/dfs/name/current/fsimage\_00000000000000000000.md5, /tmp/hadoop-  
aditya/dfs/name/current/fsimage\_00000000000000000000, /tmp/hadoop-  
aditya/dfs/name/current/seen\_txid]  
2026-05-07 17:55:37,598 INFO common.Storage: Storage directory /tmp/hadoop-  
aditya/dfs/name has been successfully formatted.  
2026-05-07 17:55:40,307 INFO namenode.FSImageFormatProtobuf: Saving image file  
/tmp/hadoop-aditya/dfs/name/current/fsimage.ckpt\_00000000000000000000 using no  
compression  
2026-05-07 17:55:40,370 INFO namenode.FSImageFormatProtobuf: Image file  
/tmp/hadoop-aditya/dfs/name/current/fsimage.ckpt\_00000000000000000000 of size 401  
bytes saved in 0 seconds .  
2026-05-07 17:55:40,378 INFO namenode.NNStorageRetentionManager: Going to retain 1  
images with txid >= 0  
2026-05-07 17:55:40,393 INFO namenode.FSNamesystem: Stopping services started for  
active state  
2026-05-07 17:55:40,394 INFO namenode.FSNamesystem: Stopping services started for  
standby state  
2026-05-07 17:55:40,397 INFO namenode.FSImage: FSImageSaver clean checkpoint:  
txid=0 when meet shutdown.  
2026-05-07 17:55:40,397 INFO namenode.NameNode: SHUTDOWN\_MSG:  
/\*\*\*\*\*  
SHUTDOWN\_MSG: Shutting down NameNode at DESKTOP-4CC1PF9/127.0.1.1  
\*\*\*\*\*/  
aditya@DESKTOP-4CC1PF9:~\$ start-dfs.sh  
Starting namenodes on [DESKTOP-4CC1PF9]  
DESKTOP-4CC1PF9: ERROR: Cannot set priority of namenode process 5494



Starting datanodes

Starting secondary namenodes [DESKTOP-4CC1PF9]

DESKTOP-4CC1PF9: ERROR: Cannot set priority of secondarynamenode process 5803

**Now type below command:-**

start-yarn.sh

**Now type:-**

jps

**If you see still see the output like below:-**

6477 Jps

6125 NodeManager

5998 ResourceManager

**Then type:-**

stop-all.sh

**Then type:-**

sudo rm -rf /tmp/hadoop-aditya

**Here “aditya” is my UNIX Username which was created when I had installed Ubuntu your UNIX Username may be different so replace “aditya” with that name in above command**

**Now type:-**

tail -n 50 /usr/local/hadoop/logs/hadoop-aditya-namenode-\*.log

**Here “aditya” is my UNIX Username which was created when I had installed Ubuntu your UNIX Username may be different so replace “aditya” with that name in above command**

**Now type:-**

nano /usr/local/hadoop/etc/hadoop/core-site.xml

**Type the below lines inside the <configuration> tags:-**

<property>

<name>fs.defaultFS</name>

<value>hdfs://localhost:9000</value>

</property>

**Now press Ctrl + O, then press Enter and then Ctrl + X**

**Now type:-**

```
sudo rm -rf /tmp/hadoop-aditya
```

**Here “aditya” is my UNIX Username which was created when I had installed Ubuntu your UNIX Username may be different so replace “aditya” with that name in above command**

**Now type:-**

```
hdfs namenode -format
```

**Now type:-**

```
start-dfs.sh
```

**And then type:-**

```
start-yarn.sh
```

**Now type:-**

```
jps
```

**You will see the output like below if the output is like below then errors are solved:-**

```
9250 Jps
8547 NameNode
8708 DataNode
8908 SecondaryNameNode
7983 NodeManager
7855 ResourceManager
```

**Step 2:-Now type this command:-**

```
start-yarn.sh
```

**Step 3:- Type the below command to make directory named “wordcount\_exam” you can give any other name as well if you want to just replace the name with wordcount\_exam**

```
mkdir wordcount_exam
```

**Step 4:-**I am using you have typed commands `start-dfs.sh` and `start-yarn.sh` and also made the directory `wordcount_exam`.Type the below command now:-

```
cd wordcount_exam
```

**Step 5:-**Now type:

```
nano input.txt
```

**And paste the below line in that file.**

```
hello world welcome to the world of big data hello
```

**Now press Ctrl + O then Enter and then Ctrl + X**

**You can type any other text as well in input.txt if you don't want to write the above text.**

**Step 6:-**Now type the below command:

```
nano mapper.py
```

**Paste the below code in the file**

```
#!/usr/bin/env python3
```

```
import sys
```

```
# Read input line by line
```

```
for line in sys.stdin:
```

```
    line = line.strip()
```

```
    words = line.split()
```

```
    for word in words:
```

```
        print(f'{word}\t1')
```

**Step 7:-**Now type the command given below:

```
nano reducer.py
```

**Paste the code given below in the file**

```
#!/usr/bin/env python3
```

```
import sys
```

```
current_word = None
current_count = 0

# Input comes from standard input
for line in sys.stdin:
    word, count = line.strip().split('\t')

    try:
        count = int(count)
    except ValueError:
        continue

    if current_word == word:
        current_count += count
    else:
        if current_word:
            print(f'{current_word}\t{current_count}')
            current_word = word
            current_count = count

# Output the last word
if current_word:
```

**Step 8:-Type the command given below:**

```
hadoop fs -mkdir -p /input
```

**Step 9:-Now type**

```
hadoop fs -put input.txt /input
```

**Step 10:-Type the command given below:-**

```
find /usr /opt ~ -name "hadoop-streaming*.jar" 2>/dev/null
```

**Output will be something like below:-**

```
/usr/local/hadoop/share/hadoop/tools/sources/hadoop-streaming-3.3.6-test-sources.jar
```

```
/usr/local/hadoop/share/hadoop/tools/sources/hadoop-streaming-3.3.6-sources.jar
```

```
/usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar
```

**So see here you will get the path and version of hadoop streaming like see in the above output /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar is hadoop streaming version and path. We will need this path in step 8 (Don't see/copy test\_sources.jar or sources.jar file remember that). And after pasting the path ensure that you type forward slash at the end “\” in the first line of the above command as you can see.**

**Step 11:-Now type**

```
hadoop jar /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar \  
-file mapper.py \  
-file reducer.py \  
-mapper "python3 mapper.py" \  
-reducer "python3 reducer.py" \  
-input input.txt \  
-output standalone_output
```

**You will see a lot of "INFO" text rapidly scrolling on your screen. Wait until it finishes completely**

**See here in this command given above the first line has will have the path and version number of hadoop streaming we got from Step No.7 so we that pasted that path after hadoop jar in the first line of the above command so your path may be little different depending on version number note this**

**Also ensure that you have named files as “input.txt”, “mapper.py” and “reducer.py” if the names are different then replace the names in the above command otherwise above command will give error**

**Step 12:-Now type:**

```
hadoop fs -cat /hdfs_output/part-00000
```

**You should see your result like given below:**

```
big 1
```

```
data 1
```

```
hello 2
```

```
of 1
```

```
the 1
```

```
to 1
```

```
welcome 1
```

```
world 2
```

**If you want to run the whole practical one more time then first type the below command:-**

```
hadoop fs -rm -r /hdfs_output
```

**And then type:-**

```
stop-all.sh
```